

1) Для заданной на схеме schema-lab3 сети, состоящей из управляемых коммутаторов и персональных компьютеров настроить на коммутаторах логическую топологию используя протокол IEEE 802.1Q, для передачи пакетов VLAN333 между коммутаторами использовать Native VLAN

Подключение ПК:

Коммутатор 3:

```
Switch#enable
```

```
Switch#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#int gi1/0
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 20
```

```
Switch(config-if)#exit
```

```
Switch(config)#int gi1/1
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 333
```

```
Switch(config-if)#exit
```

```
Switch(config)#exit
```

Коммутатор 4:

```
Switch>enable
```

```
Switch#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#int gi1/0
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 20
```

```
Switch(config-if)#exit  
Switch(config)#int gi1/1  
Switch(config-if)#switchport mode access  
Switch(config-if)#switchport access vlan 333  
Switch(config-if)#exit  
Switch(config)#exit
```

Коммутатор 5:

```
Switch>enable  
Switch#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
Switch(config)#int gi1/0  
Switch(config-if)#switchport mode access  
Switch(config-if)#switchport access vlan 20  
Switch(config-if)#exit  
Switch(config)#int gi1/1  
Switch(config-if)#switchport mode access  
Switch(config-if)#switchport access vlan 333  
Switch(config-if)#exit  
Switch(config)#exit
```

Настройка VLAN:

Коммутатор 1:

```
Switch#enable  
Switch#conf t  
Enter configuration commands, one per line. End with CNTL/Z.
```

```
Switch(config)#vlan 123
Switch(config-vlan)#exit
Switch(config)#int gi0/0
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 20
Switch(config-if)#exit
Switch(config)#int gi0/1
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 333
Switch(config-if)#switchport trunk allowed vlan 333
Switch(config-if)#exit
Switch(config)#int gi0/2
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 20
Switch(config-if)#exit
Switch(config)#int gi0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 123
Switch(config-if)#exit
Switch(config)#int gi1/0
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 20
Switch(config-if)#exit
Switch(config)#int gi1/1
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 123
Switch(config-if)#exit
Switch(config)#int gi1/2
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 20
Switch(config-if)#exit
Switch(config)#int gi1/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 123
Switch(config-if)#exit
Switch(config)#exit
```

Коммутатор 2:

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 123
Switch(config-vlan)#exit
Switch(config)#int gi0/0
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 20
Switch(config-if)#exit
Switch(config)#int gi0/1
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 333
```

```
Switch(config-if)#switchport trunk allowed vlan 333
Switch(config-if)#exit
Switch(config)#int gi0/2
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 333
Switch(config-if)#switchport trunk allowed vlan 333
Switch(config-if)#exit
Switch(config)#int gi0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 123
Switch(config-if)#exit
Switch(config)#int gi1/0
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 333
Switch(config-if)#switchport trunk allowed vlan 333
Switch(config-if)#exit
Switch(config)#int gi1/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 123
Switch(config-if)#exit
Switch(config)#int gi1/2
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 333
Switch(config-if)#switchport trunk allowed vlan 333
Switch(config-if)#exit
Switch(config)#int gi1/3
```

```
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 123
Switch(config-if)#exit
Switch(config)#exit
```

Коммутатор 3:

```
Switch>enable
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int gi0/0
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 20
Switch(config-if)#exit
Switch(config)#int gi0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 123
% Access VLAN does not exist. Creating vlan 123
Switch(config-if)#exit
Switch(config)#int gi0/2
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk native vlan 333
Switch(config-if)#switchport trunk allowed vlan 333
Switch(config-if)#exit
Switch(config)#int gi0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 123
```

Switch(config-if)#exit

Switch(config)#exit

Коммутатор 4:

Switch>enable

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#vlan 123

Switch(config-vlan)#exit

Switch(config)#int gi0/0

Switch(config-if)#switchport trunk encapsulation dot1q

Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk allowed vlan 20

Switch(config-if)#exit

Switch(config)#int gi0/1

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 123

Switch(config-if)#exit

Switch(config)#int gi0/2

Switch(config-if)#switchport trunk encapsulation dot1q

Switch(config-if)#switchport mode trunk

Switch(config-if)#switchport trunk native vlan 333

Switch(config-if)#switchport trunk allowed vlan 333

Switch(config-if)#exit

Switch(config)#int gi0/3

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 123

Switch(config-if)#exit

```
Switch(config)#exit
```

Коммутатор 5:

```
Switch>enable
```

```
Switch#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#vlan 123
```

```
Switch(config-vlan)#exit
```

```
Switch(config)#int gi0/0
```

```
Switch(config-if)#switchport trunk encapsulation dot1q
```

```
Switch(config-if)#switchport mode trunk
```

```
Switch(config-if)#switchport trunk allowed vlan 20
```

```
Switch(config-if)#exit
```

```
Switch(config)#int gi0/1
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 123
```

```
Switch(config-if)#exit
```

```
Switch(config)#int gi0/2
```

```
Switch(config-if)#switchport trunk encapsulation dot1q
```

```
Switch(config-if)#switchport mode trunk
```

```
Switch(config-if)#switchport trunk native vlan 333
```

```
Switch(config-if)#switchport trunk allowed vlan 333
```

```
Switch(config-if)#exit
```

```
Switch(config)#int gi0/3
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 123
```

```
Switch(config-if)#exit
```

```
Switch(config)#exit
```


2) Проверить доступность персональных компьютеров, находящихся в одинаковых VLAN и недоступность находящихся в различных, результаты задокументировать

Назначение ip:

PC1:

PC1> ip 192.168.20.1/24

Checking for duplicate address...

PC1 : 192.168.20.1 255.255.255.0

PC2:

PC2> ip 192.168.20.2/24

Checking for duplicate address...

PC2 : 192.168.20.2 255.255.255.0

PC3:

PC3> ip 192.168.20.3/24

Checking for duplicate address...

PC3 : 192.168.20.3 255.255.255.0

PC4:

PC4> ip 192.168.20.4/24

Checking for duplicate address...

PC4 : 192.168.20.4 255.255.255.0

PC5:

PC5> ip 192.168.20.5/24

Checking for duplicate address...

PC5 : 192.168.20.5 255.255.255.0

PC6:

PC6> ip 192.168.20.6/24

Checking for duplicate address...

PC6 : 192.168.20.6 255.255.255.0

On PC1:

PC1> ping 192.168.20.3

84 bytes from 192.168.20.3 icmp_seq=1 ttl=64 time=16.038 ms

84 bytes from 192.168.20.3 icmp_seq=2 ttl=64 time=5.510 ms

84 bytes from 192.168.20.3 icmp_seq=3 ttl=64 time=4.259 ms

84 bytes from 192.168.20.3 icmp_seq=4 ttl=64 time=8.108 ms

84 bytes from 192.168.20.3 icmp_seq=5 ttl=64 time=3.251 ms

PC1> ping 192.168.20.5

84 bytes from 192.168.20.5 icmp_seq=1 ttl=64 time=16.902 ms

84 bytes from 192.168.20.5 icmp_seq=2 ttl=64 time=6.581 ms

84 bytes from 192.168.20.5 icmp_seq=3 ttl=64 time=12.016 ms

84 bytes from 192.168.20.5 icmp_seq=4 ttl=64 time=9.083 ms

84 bytes from 192.168.20.5 icmp_seq=5 ttl=64 time=8.398 ms

PC1> ping 192.168.20.2

host (192.168.20.2) not reachable

PC1> ping 192.168.20.4

host (192.168.20.4) not reachable

PC1> ping 192.168.20.6

host (192.168.20.6) not reachable

От PC2:

PC2> ping 192.168.20.4

84 bytes from 192.168.20.4 icmp_seq=1 ttl=64 time=19.245 ms

84 bytes from 192.168.20.4 icmp_seq=2 ttl=64 time=2.866 ms

84 bytes from 192.168.20.4 icmp_seq=3 ttl=64 time=6.137 ms

84 bytes from 192.168.20.4 icmp_seq=4 ttl=64 time=6.849 ms

84 bytes from 192.168.20.4 icmp_seq=5 ttl=64 time=4.410 ms

PC2> ping 192.168.20.6

84 bytes from 192.168.20.6 icmp_seq=1 ttl=64 time=13.783 ms

84 bytes from 192.168.20.6 icmp_seq=2 ttl=64 time=5.401 ms

84 bytes from 192.168.20.6 icmp_seq=3 ttl=64 time=8.026 ms

84 bytes from 192.168.20.6 icmp_seq=4 ttl=64 time=9.832 ms

84 bytes from 192.168.20.6 icmp_seq=5 ttl=64 time=12.863 ms

PC2> ping 192.168.20.1

host (192.168.20.1) not reachable

PC2> ping 192.168.20.3

host (192.168.20.3) not reachable

PC2> ping 192.168.20.5

host (192.168.20.5) not reachable

On PC3:

PC3> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=64 time=9.893 ms

84 bytes from 192.168.20.1 icmp_seq=2 ttl=64 time=4.108 ms

84 bytes from 192.168.20.1 icmp_seq=3 ttl=64 time=9.097 ms

84 bytes from 192.168.20.1 icmp_seq=4 ttl=64 time=10.042 ms

84 bytes from 192.168.20.1 icmp_seq=5 ttl=64 time=7.457 ms

PC3> ping 192.168.20.5

84 bytes from 192.168.20.5 icmp_seq=1 ttl=64 time=9.031 ms

84 bytes from 192.168.20.5 icmp_seq=2 ttl=64 time=3.372 ms

84 bytes from 192.168.20.5 icmp_seq=3 ttl=64 time=3.856 ms

84 bytes from 192.168.20.5 icmp_seq=4 ttl=64 time=7.908 ms

84 bytes from 192.168.20.5 icmp_seq=5 ttl=64 time=6.053 ms

PC3> ping 192.168.20.2

host (192.168.20.2) not reachable

PC3> ping 192.168.20.4

host (192.168.20.4) not reachable

PC3> ping 192.168.20.6

host (192.168.20.6) not reachable

От PC4:

PC4> ping 192.168.20.2

84 bytes from 192.168.20.2 icmp_seq=1 ttl=64 time=17.995 ms

84 bytes from 192.168.20.2 icmp_seq=2 ttl=64 time=3.477 ms

84 bytes from 192.168.20.2 icmp_seq=3 ttl=64 time=4.844 ms

84 bytes from 192.168.20.2 icmp_seq=4 ttl=64 time=5.839 ms

84 bytes from 192.168.20.2 icmp_seq=5 ttl=64 time=4.669 ms

PC4> ping 192.168.20.6

84 bytes from 192.168.20.6 icmp_seq=1 ttl=64 time=6.404 ms

84 bytes from 192.168.20.6 icmp_seq=2 ttl=64 time=2.173 ms

84 bytes from 192.168.20.6 icmp_seq=3 ttl=64 time=9.054 ms

84 bytes from 192.168.20.6 icmp_seq=4 ttl=64 time=11.591 ms

84 bytes from 192.168.20.6 icmp_seq=5 ttl=64 time=8.657 ms

PC4> ping 192.168.20.1

host (192.168.20.1) not reachable

PC4> ping 192.168.20.3

host (192.168.20.3) not reachable

PC4> ping 192.168.20.5

host (192.168.20.5) not reachable

Or PC5:

PC5> ping 192.168.20.1

84 bytes from 192.168.20.1 icmp_seq=1 ttl=64 time=2.893 ms

84 bytes from 192.168.20.1 icmp_seq=2 ttl=64 time=7.514 ms

84 bytes from 192.168.20.1 icmp_seq=3 ttl=64 time=5.826 ms

84 bytes from 192.168.20.1 icmp_seq=4 ttl=64 time=8.070 ms

84 bytes from 192.168.20.1 icmp_seq=5 ttl=64 time=9.465 ms

PC5> ping 192.168.20.3

84 bytes from 192.168.20.3 icmp_seq=1 ttl=64 time=10.207 ms

84 bytes from 192.168.20.3 icmp_seq=2 ttl=64 time=5.693 ms

84 bytes from 192.168.20.3 icmp_seq=3 ttl=64 time=9.358 ms

84 bytes from 192.168.20.3 icmp_seq=4 ttl=64 time=2.551 ms

84 bytes from 192.168.20.3 icmp_seq=5 ttl=64 time=8.489 ms

PC5> ping 192.168.20.2

host (192.168.20.2) not reachable

PC5> ping 192.168.20.4

host (192.168.20.4) not reachable

PC5> ping 192.168.20.6

host (192.168.20.6) not reachable

Or PC6:

PC6> ping 192.168.20.2

84 bytes from 192.168.20.2 icmp_seq=1 ttl=64 time=11.306 ms

84 bytes from 192.168.20.2 icmp_seq=2 ttl=64 time=8.506 ms

84 bytes from 192.168.20.2 icmp_seq=3 ttl=64 time=6.467 ms

84 bytes from 192.168.20.2 icmp_seq=4 ttl=64 time=11.673 ms

84 bytes from 192.168.20.2 icmp_seq=5 ttl=64 time=15.880 ms

PC6> ping 192.168.20.4

84 bytes from 192.168.20.4 icmp_seq=1 ttl=64 time=7.652 ms

84 bytes from 192.168.20.4 icmp_seq=2 ttl=64 time=6.712 ms

84 bytes from 192.168.20.4 icmp_seq=3 ttl=64 time=8.072 ms

84 bytes from 192.168.20.4 icmp_seq=4 ttl=64 time=4.385 ms

84 bytes from 192.168.20.4 icmp_seq=5 ttl=64 time=10.165 ms

PC6> ping 192.168.20.1

host (192.168.20.1) not reachable

PC6> ping 192.168.20.3

host (192.168.20.3) not reachable

PC6> ping 192.168.20.5

host (192.168.20.5) not reachable

3) Перехватить в WireShark пакеты с тегами и без тегов (nb!), результаты задокументировать

От PC1 к PC3:

Между компьютерами и коммутаторами пакеты передаются без тегов:

Захват с Standard input [Layer2Switch-3 Ethernet4 to PC1 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
65	76.241789	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
66	78.250858	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
67	80.254858	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
68	81.909211	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.20.3? Tell 192.168.20.1
69	81.915805	Private_66:68:02	Private_66:68:00	ARP	64	192.168.20.3 is at 00:50:79:66:68:02
70	81.916562	192.168.20.1	192.168.20.3	ICMP	98	Echo (ping) request id=0xfdd2, seq=1/256, ttl=64 (reply in 71)
71	81.931915	192.168.20.3	192.168.20.1	ICMP	98	Echo (ping) reply id=0xfdd2, seq=1/256, ttl=64 (request in 70)
72	82.214274	0c:fa:40:2e:00:04	0c:fa:40:2e:00:04	LOOP	60	Reply
73	82.258756	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
74	82.932502	192.168.20.1	192.168.20.3	ICMP	98	Echo (ping) request id=0xfed2, seq=2/512, ttl=64 (reply in 75)
75	82.941583	192.168.20.3	192.168.20.1	ICMP	98	Echo (ping) reply id=0xfed2, seq=2/512, ttl=64 (request in 74)
76	83.942125	192.168.20.1	192.168.20.3	ICMP	98	Echo (ping) request id=0xffd2, seq=3/768, ttl=64 (reply in 77)
77	83.947943	192.168.20.3	192.168.20.1	ICMP	98	Echo (ping) reply id=0xffd2, seq=3/768, ttl=64 (request in 76)
78	84.269430	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
79	84.948570	192.168.20.1	192.168.20.3	ICMP	98	Echo (ping) request id=0x00d3, seq=4/1024, ttl=64 (reply in 80)
80	84.958811	192.168.20.3	192.168.20.1	ICMP	98	Echo (ping) reply id=0x00d3, seq=4/1024, ttl=64 (request in 79)
81	85.959150	192.168.20.1	192.168.20.3	ICMP	98	Echo (ping) request id=0x01d3, seq=5/1280, ttl=64 (reply in 82)
82	85.976078	192.168.20.3	192.168.20.1	ICMP	98	Echo (ping) reply id=0x01d3, seq=5/1280, ttl=64 (request in 81)
83	86.275237	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
84	86.832987	0c:fa:40:2e:00:04	CDP/VTP/DTP/PagP/UD...	DTP	60	Dynamic Trunk Protocol
85	86.833024	0c:fa:40:2e:00:04	CDP/VTP/DTP/PagP/UD...	DTP	90	Dynamic Trunk Protocol
86	88.278117	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
87	90.284622	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
88	92.228360	0c:fa:40:2e:00:04	0c:fa:40:2e:00:04	LOOP	60	Reply

> Frame 70: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:02 (00:50:79:66:68:02)

> Internet Protocol Version 4, Src: 192.168.20.1, Dst: 192.168.20.3

> Internet Control Message Protocol

0000 00 50
0010 00 54
0020 14 03
0030 0e 0f
0040 1e 1f
0050 2e 2f

Захват с Standard input [Layer2Switch-4 Ethernet4 to PC3 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
51	55.064441	0c:6b:9e:d5:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
52	57.066808	0c:6b:9e:d5:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
53	59.075506	0c:6b:9e:d5:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
54	60.054896	0c:6b:9e:d5:00:04	0c:6b:9e:d5:00:04	LOOP	60	Reply
55	61.079368	0c:6b:9e:d5:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
56	62.735978	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.20.3? Tell 192.168.20.1
57	62.736079	Private_66:68:02	Private_66:68:00	ARP	64	192.168.20.3 is at 00:50:79:66:68:02
58	62.748077	192.168.20.1	192.168.20.3	ICMP	98	Echo (ping) request id=0xfdd2, seq=1/256, ttl=64 (reply in 59)
59	62.748170	192.168.20.3	192.168.20.1	ICMP	98	Echo (ping) reply id=0xfdd2, seq=1/256, ttl=64 (request in 58)
60	63.083781	0c:6b:9e:d5:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
61	63.759107	192.168.20.1	192.168.20.3	ICMP	98	Echo (ping) request id=0xfed2, seq=2/512, ttl=64 (reply in 62)
62	63.759304	192.168.20.3	192.168.20.1	ICMP	98	Echo (ping) reply id=0xfed2, seq=2/512, ttl=64 (request in 61)
63	64.769538	192.168.20.1	192.168.20.3	ICMP	98	Echo (ping) request id=0xffd2, seq=3/768, ttl=64 (reply in 64)
64	64.769678	192.168.20.3	192.168.20.1	ICMP	98	Echo (ping) reply id=0xffd2, seq=3/768, ttl=64 (request in 63)
65	65.093577	0c:6b:9e:d5:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
66	65.774831	192.168.20.1	192.168.20.3	ICMP	98	Echo (ping) request id=0x00d3, seq=4/1024, ttl=64 (reply in 67)
67	65.774935	192.168.20.3	192.168.20.1	ICMP	98	Echo (ping) reply id=0x00d3, seq=4/1024, ttl=64 (request in 66)
68	66.791841	192.168.20.1	192.168.20.3	ICMP	98	Echo (ping) request id=0x01d3, seq=5/1280, ttl=64 (reply in 69)
69	66.791957	192.168.20.3	192.168.20.1	ICMP	98	Echo (ping) reply id=0x01d3, seq=5/1280, ttl=64 (request in 68)
70	67.100415	0c:6b:9e:d5:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
71	69.102775	0c:6b:9e:d5:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
72	69.496178	0c:6b:9e:d5:00:04	CDP/VTP/DTP/PagP/UD...	CDP	424	Device ID: Switch Port ID: GigabitEthernet1/0
73	70.074377	0c:6b:9e:d5:00:04	0c:6b:9e:d5:00:04	LOOP	60	Reply
74	71.110682	0c:6b:9e:d5:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005

> Frame 58: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:02 (00:50:79:66:68:02)

> Internet Protocol Version 4, Src: 192.168.20.1, Dst: 192.168.20.3

> Internet Control Message Protocol

0000 00 5
0010 00 5
0020 14 0
0030 0e 0
0040 1e 1
0050 2e 2

Между коммутаторами пакеты передаются с тегом:

Захват с Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка



Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
77	80.193671	0c:fa:40:2e:00:00	0c:fa:40:2e:00:00	LOOP	60	Reply
78	80.257775	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
79	82.261505	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
80	84.264170	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
81	86.272893	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
82	87.578563	0c:5f:1e:17:00:02	0c:5f:1e:17:00:02	LOOP	60	Reply
83	88.276585	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
84	89.932764	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.3? Tell 192.168.20.1
85	89.937113	Private_66:68:02	Private_66:68:00	ARP	68	192.168.20.3 is at 00:50:79:66:68:02
86	89.945062	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0xfdd2, seq=1/256, ttl=64 (reply in 87)
87	89.954692	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0xfdd2, seq=1/256, ttl=64 (request in 86)
88	90.199758	0c:fa:40:2e:00:00	0c:fa:40:2e:00:00	LOOP	60	Reply
89	90.281207	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
90	90.956241	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0xfed2, seq=2/512, ttl=64 (reply in 91)
91	90.962652	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0xfed2, seq=2/512, ttl=64 (request in 90)
92	91.966557	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0xffd2, seq=3/768, ttl=64 (reply in 93)
93	91.970659	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0xffd2, seq=3/768, ttl=64 (request in 92)
94	92.290990	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
95	92.972004	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0x00d3, seq=4/1024, ttl=64 (reply in 96)
96	92.981504	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0x00d3, seq=4/1024, ttl=64 (request in 95)
97	93.982531	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0x01d3, seq=5/1280, ttl=64 (reply in 98)
98	93.998663	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0x01d3, seq=5/1280, ttl=64 (request in 97)
99	94.297629	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
100	96.300296	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003

> Frame 86: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:02 (00:50:79:66:68:02)
▼ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
 000. = Priority: Best Effort (default) (0)
 ...0 = DEI: Ineligible
 0000 0001 0100 = ID: 20
 Type: IPv4 (0x0800)
> Internet Protocol Version 4, Src: 192.168.20.1, Dst: 192.168.20.3
> Internet Control Message Protocol

0000 00
0010 08
0020 14
0030 0a
0040 1a
0050 2a
0060 3a

Захват с Standard input [Layer2Switch-1 Ethernet4 to Layer2Switch-4 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка



Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
47	42.132542	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
48	44.141291	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
49	45.119127	0c:6b:9e:d5:00:00	0c:6b:9e:d5:00:00	LOOP	60	Reply
50	45.446974	0c:5f:1e:17:00:04	0c:5f:1e:17:00:04	LOOP	60	Reply
51	46.144964	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
52	47.304157	0c:6b:9e:d5:00:00	CDP/VTP/DTP/PagP/UD...	CDP	424	Device ID: Switch Port ID: GigabitEthernet0/0
53	47.801939	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.3? Tell 192.168.20.1
54	47.805010	Private_66:68:02	Private_66:68:00	ARP	68	192.168.20.3 is at 00:50:79:66:68:02
55	47.813945	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0xfdd2, seq=1/256, ttl=64 (reply in 56)
56	47.816854	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0xfdd2, seq=1/256, ttl=64 (request in 55)
57	48.149590	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
58	48.824972	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0xfed2, seq=2/512, ttl=64 (reply in 59)
59	48.830638	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0xfed2, seq=2/512, ttl=64 (request in 58)
60	49.835583	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0xffd2, seq=3/768, ttl=64 (reply in 61)
61	49.838515	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0xffd2, seq=3/768, ttl=64 (request in 60)
62	50.159412	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
63	50.840870	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0x00d3, seq=4/1024, ttl=64 (reply in 64)
64	50.849326	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0x00d3, seq=4/1024, ttl=64 (request in 63)
65	51.858056	192.168.20.1	192.168.20.3	ICMP	102	Echo (ping) request id=0x01d3, seq=5/1280, ttl=64 (reply in 66)
66	51.858648	192.168.20.3	192.168.20.1	ICMP	102	Echo (ping) reply id=0x01d3, seq=5/1280, ttl=64 (request in 65)
67	52.166003	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
68	54.168665	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
69	55.138570	0c:6b:9e:d5:00:00	0c:6b:9e:d5:00:00	LOOP	60	Reply
70	55.475529	0c:5f:1e:17:00:04	0c:5f:1e:17:00:04	LOOP	60	Reply

> Frame 55: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:02 (00:50:79:66:68:02)
▼ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
 000. = Priority: Best Effort (default) (0)
 ...0 = DEI: Ineligible
 0000 0001 0100 = ID: 20
 Type: IPv4 (0x0800)
> Internet Protocol Version 4, Src: 192.168.20.1, Dst: 192.168.20.3
> Internet Control Message Protocol

0000 00 50
0010 08 00
0020 14 01
0030 0a 0b
0040 1a 1b
0050 2a 2b
0060 3a 3b

При пинге компьютера из другого VLAN (PC4) ARP-запрос уходит к коммутатору без тега:

Захват с Standard input [Layer2Switch-3 Ethernet4 to PC1 Ethernet0]

ФайлПравкаВидЗапускЗахватАнализСтатистикаТелефонияБеспроводная связьИнструментыСправка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
518	716.168654	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
519	717.392093	0c:fa:40:2e:00:04	CDP/VTP/DTP/PAgP/UD...	DTP	60	Dynamic Trunk Protocol
520	717.392147	0c:fa:40:2e:00:04	CDP/VTP/DTP/PAgP/UD...	DTP	90	Dynamic Trunk Protocol
521	718.173257	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
522	720.179607	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
523	722.184807	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
524	722.784538	0c:fa:40:2e:00:04	0c:fa:40:2e:00:04	LOOP	60	Reply
525	724.194389	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
526	726.196991	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
527	728.201491	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
528	730.206603	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
529	730.781182	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.20.4? Tell 192.168.20.1
530	731.781340	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.20.4? Tell 192.168.20.1
531	732.208284	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
532	732.781408	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.20.4? Tell 192.168.20.1
533	732.791988	0c:fa:40:2e:00:04	0c:fa:40:2e:00:04	LOOP	60	Reply
534	734.220195	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
535	736.223586	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
536	738.224469	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
537	740.225884	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
538	742.228515	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
539	742.800653	0c:fa:40:2e:00:04	0c:fa:40:2e:00:04	LOOP	60	Reply
540	744.238035	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
541	746.243042	0c:fa:40:2e:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005

> Frame 529: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0

> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

> Address Resolution Protocol (request)

0000ff ff ff

001008 00 06

0020ff ff ff

003000 00 00

Между коммутаторами перемещается с тегом:

Захват с Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка



Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
615	730.206997	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
616	730.771412	0c:fa:40:2e:00:00	0c:fa:40:2e:00:00	LOOP	60	Reply
617	732.215668	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
618	734.219375	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
619	736.223103	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
620	738.228778	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
621	738.804715	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.4? Tell 192.168.20.1
622	739.442486	0c:5f:1e:17:00:02	0c:5f:1e:17:00:02	LOOP	60	Reply
623	739.804885	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.4? Tell 192.168.20.1
624	740.230472	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
625	740.778923	0c:fa:40:2e:00:00	0c:fa:40:2e:00:00	LOOP	60	Reply
626	740.804844	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.4? Tell 192.168.20.1
627	741.860166	0c:5f:1e:17:00:02	CDP/VTP/DTP/PagP/UD...	CDP	424	Device ID: Switch Port ID: GigabitEthernet0/2
628	741.999496	0c:5f:1e:17:00:02	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol
629	742.242158	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
630	744.245895	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
631	744.831294	0c:fa:40:2e:00:00	CDP/VTP/DTP/PagP/UD...	CDP	424	Device ID: Switch Port ID: GigabitEthernet0/0
632	746.246808	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
633	748.247223	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
634	749.461963	0c:5f:1e:17:00:02	0c:5f:1e:17:00:02	LOOP	60	Reply
635	750.250911	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
636	750.787361	0c:fa:40:2e:00:00	0c:fa:40:2e:00:00	LOOP	60	Reply
637	752.259673	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003
638	754.265380	0c:5f:1e:17:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8003

> Frame 621: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▼ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
0000 = Priority: Best Effort (default) (0)
...0 = DEI: Ineligible
.... 0000 0001 0100 = ID: 20
Type: ARP (0x0806)
Padding: 00
Trailer: 00000000
> Address Resolution Protocol (request)

0000 ff
0010 08
0020 c0
0030 00
0040 00

Захват с Standard input [Layer2Switch-1 Ethernet4 to Layer2Switch-4 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка



Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
577	682.059303	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
578	683.147125	0c:6b:9e:d5:00:00	CDP/VTP/DTP/PagP/UD...	CDP	424	Device ID: Switch Port ID: GigabitEthernet0/0
579	684.063907	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
580	685.481006	0c:6b:9e:d5:00:00	0c:6b:9e:d5:00:00	LOOP	60	Reply
581	686.069677	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
582	687.285546	0c:5f:1e:17:00:04	0c:5f:1e:17:00:04	LOOP	60	Reply
583	688.075366	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
584	690.084029	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
585	692.087758	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
586	694.091485	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
587	695.485256	0c:6b:9e:d5:00:00	0c:6b:9e:d5:00:00	LOOP	60	Reply
588	696.097125	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
589	696.673761	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.4? Tell 192.168.20.1
590	697.310903	0c:5f:1e:17:00:04	0c:5f:1e:17:00:04	LOOP	60	Reply
591	697.673764	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.4? Tell 192.168.20.1
592	698.098851	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
593	698.673876	Private_66:68:00	Broadcast	ARP	68	Who has 192.168.20.4? Tell 192.168.20.1
594	699.868809	0c:5f:1e:17:00:04	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol
595	700.110584	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
596	701.224381	0c:6b:9e:d5:00:00	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol
597	702.114267	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
598	704.115289	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005
599	705.486092	0c:6b:9e:d5:00:00	0c:6b:9e:d5:00:00	LOOP	60	Reply
600	706.115625	0c:5f:1e:17:00:04	PVST+	STP	68	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8005

> Frame 589: Packet, 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface -, id 0
> Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
▼ 802.1Q Virtual LAN, PRI: 0, DEI: 0, ID: 20
0000 = Priority: Best Effort (default) (0)
...0 = DEI: Ineligible
.... 0000 0001 0100 = ID: 20
Type: ARP (0x0806)
Padding: 00
Trailer: 00000000
> Address Resolution Protocol (request)

0000 ff
0010 08
0020 c0
0030 00
0040 00

Захват с Standard input [Layer2Switch-4 Ethernet5 to PC4 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты

arp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	192.168.1.1	192.168.1.254	ARP	60	Request from 192.168.1.1 to 192.168.1.254

Между компьютерами и коммутаторами пакеты также передаются без тегов:

The figure displays a Wireshark network traffic capture. The top pane shows the packet list with columns for No., Time, Source, Destination, Protocol, Length, and Info. Packet 69 is highlighted, showing an ARP Request from Private 66:68:01 to Broadcast. The bottom pane shows the packet details for the selected packet, indicating it is an Ethernet II frame from Private 66:68:01 to Broadcast (ff:ff:ff:ff:ff:ff) over interface -.

No.	Time	Source	Destination	Protocol	Length	Info
64	92.268412	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
65	94.276505	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
66	96.274408	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
67	96.712884	0c:fa:40:2e:00:05	0c:fa:40:2e:00:05	LOOP	60	Reply
68	98.285381	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
69	100.292558	Private 66:68:01	Broadcast	ARP	64	Who has 192.168.20.4? Tell 192.168.20.2
70	100.300094	Private 66:68:03	Private 66:68:01	ARP	64	192.168.20.4 is at 00:50:79:66:68:03
71	100.300780	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
72	100.301115	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6ad9, seq=1/256, ttl=64 (reply in 73)
73	100.310496	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6ad9, seq=1/256, ttl=64 (request in 72)
74	101.310728	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6bd9, seq=2/512, ttl=64 (reply in 75)
75	101.328116	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6bd9, seq=2/512, ttl=64 (request in 74)
76	102.301570	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
77	102.328620	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6cd9, seq=3/768, ttl=64 (reply in 78)
78	102.337862	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6cd9, seq=3/768, ttl=64 (request in 77)
79	103.338432	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6dd9, seq=4/1024, ttl=64 (reply in 80)
80	103.343824	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6dd9, seq=4/1024, ttl=64 (request in 79)
81	104.312436	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
82	104.344911	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6ed9, seq=5/1280, ttl=64 (reply in 83)
83	104.347525	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6ed9, seq=5/1280, ttl=64 (request in 82)
84	104.409408	0c:fa:40:2e:00:05	CDP/VTP/DTP/PagP/UD...	CDP	424	Device ID: Switch Port ID: GigabitEthernet1/1
85	106.312382	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
86	106.724388	0c:fa:40:2e:00:05	0c:fa:40:2e:00:05	LOOP	60	Reply
87	108.315540	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006

> Frame 69: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0	0000 ff f
> Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)	0010 08 0
> Address Resolution Protocol (request)	0020 ff f
	0030 00 0

Захват с Standard input [Layer2Switch-4 Ethernet5 to PC4 Ethernet0]						
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка						
Примените фильтр отображения ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
62	87.414515	0c:6b:9e:d5:00:05	0c:6b:9e:d5:00:05	LOOP	60	Reply
63	88.370875	0c:6b:9e:d5:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
64	90.047147	0c:6b:9e:d5:00:05	CDP/VTP/DTP/PagP/UD...	DTP	60	Dynamic Trunk Protocol
65	90.047189	0c:6b:9e:d5:00:05	CDP/VTP/DTP/PagP/UD...	DTP	90	Dynamic Trunk Protocol
66	90.373121	0c:6b:9e:d5:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
67	92.383035	0c:6b:9e:d5:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
68	94.392492	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.4? Tell 192.168.20.2
69	94.392571	Private_66:68:03	Private_66:68:01	ARP	64	192.168.20.4 is at 00:50:79:66:68:03
70	94.396134	0c:6b:9e:d5:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
71	94.407800	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6ad9, seq=1/256, ttl=64 (reply in 72)
72	94.407936	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6ad9, seq=1/256, ttl=64 (request in 71)
73	95.416655	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6bd9, seq=2/512, ttl=64 (reply in 74)
74	95.416771	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6bd9, seq=2/512, ttl=64 (request in 73)
75	96.400350	0c:6b:9e:d5:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
76	96.428542	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6cd9, seq=3/768, ttl=64 (reply in 77)
77	96.428676	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6cd9, seq=3/768, ttl=64 (request in 76)
78	97.424310	0c:6b:9e:d5:00:05	0c:6b:9e:d5:00:05	LOOP	60	Reply
79	97.439602	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6dd9, seq=4/1024, ttl=64 (reply in 80)
80	97.439800	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6dd9, seq=4/1024, ttl=64 (request in 79)
81	98.411725	0c:6b:9e:d5:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
82	98.444685	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6ed9, seq=5/1280, ttl=64 (reply in 83)
83	98.444822	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6ed9, seq=5/1280, ttl=64 (request in 82)
84	100.411266	0c:6b:9e:d5:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
85	102.414602	0c:6b:9e:d5:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
> Frame 68: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0 > Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff) > Address Resolution Protocol (request)						
						0000 ff f 0010 08 0 0020 ff f 0030 00 0

Между коммутаторами пакеты передаются теперь без тега:

Захват с Standard input [Layer2Switch-3 Ethernet2 to Layer2Switch-2 Ethernet2]						
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка						
Примените фильтр отображения ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
61	72.206121	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8003
62	74.209585	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8003
63	76.212213	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8003
64	76.647981	0c:fa:40:2e:00:02	0c:fa:40:2e:00:02	LOOP	60	Reply
65	76.658174	0c:88:ae:68:00:02	0c:88:ae:68:00:02	LOOP	60	Reply
66	78.221871	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8003
67	80.231349	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.4? Tell 192.168.20.2
68	80.234972	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8003
69	80.236578	Private_66:68:03	Private_66:68:01	ARP	64	192.168.20.4 is at 00:50:79:66:68:03
70	80.246585	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6ad9, seq=1/256, ttl=64 (reply in 71)
71	80.248625	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6ad9, seq=1/256, ttl=64 (request in 70)
72	81.255533	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6bd9, seq=2/512, ttl=64 (reply in 73)
73	81.266126	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6bd9, seq=2/512, ttl=64 (request in 72)
74	82.239271	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8003
75	82.267447	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6cd9, seq=3/768, ttl=64 (reply in 76)
76	82.275944	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6cd9, seq=3/768, ttl=64 (request in 75)
77	83.277258	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6dd9, seq=4/1024, ttl=64 (reply in 78)
78	83.280779	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6dd9, seq=4/1024, ttl=64 (request in 77)
79	83.510358	0c:fa:40:2e:00:02	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol
80	84.250259	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8003
81	84.283842	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6ed9, seq=5/1280, ttl=64 (reply in 82)
82	84.285810	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6ed9, seq=5/1280, ttl=64 (request in 81)
83	86.249895	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8003
84	86.659478	0c:fa:40:2e:00:02	0c:fa:40:2e:00:02	LOOP	60	Reply
> Frame 67: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0 > Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff) > Address Resolution Protocol (request)						
						0000 ff ff 0010 08 00 0020 ff ff 0030 00 00

Захват с Standard input [Layer2Switch-2 Ethernet4 to Layer2Switch-4 Ethernet2]						
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка						
Примените фильтр отображения ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
4	3.049422	0c:6b:9e:d5:00:02	0c:6b:9e:d5:00:02	LOOP	60	Reply
5	4.007459	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
6	6.010068	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
7	6.457543	0c:88:ae:68:00:04	0c:88:ae:68:00:04	LOOP	60	Reply
8	8.019735	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
9	10.029650	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.4? Tell 192.168.20.2
10	10.032830	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
11	10.033449	Private_66:68:03	Private_66:68:01	ARP	64	192.168.20.4 is at 00:50:79:66:68:03
12	10.045094	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6ad9, seq=1/256, ttl=64 (reply in 13)
13	10.045783	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6ad9, seq=1/256, ttl=64 (request in 12)
14	10.952910	0c:88:ae:68:00:04	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol
15	11.053838	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6bd9, seq=2/512, ttl=64 (reply in 16)
16	11.056708	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6bd9, seq=2/512, ttl=64 (request in 15)
17	12.037131	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
18	12.065604	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6cd9, seq=3/768, ttl=64 (reply in 19)
19	12.072563	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6cd9, seq=3/768, ttl=64 (request in 18)
20	13.058588	0c:6b:9e:d5:00:02	0c:6b:9e:d5:00:02	LOOP	60	Reply
21	13.076842	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6dd9, seq=4/1024, ttl=64 (reply in 22)
22	13.077716	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6dd9, seq=4/1024, ttl=64 (request in 21)
23	14.048203	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
24	14.081896	192.168.20.2	192.168.20.4	ICMP	98	Echo (ping) request id=0x6ed9, seq=5/1280, ttl=64 (reply in 25)
25	14.083239	192.168.20.4	192.168.20.2	ICMP	98	Echo (ping) reply id=0x6ed9, seq=5/1280, ttl=64 (request in 24)
26	16.047778	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8005
27	16.498702	0c:88:ae:68:00:04	0c:88:ae:68:00:04	LOOP	60	Reply
> Frame 9: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0 > Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff) > Address Resolution Protocol (request)						
					0000	ff f
					0010	08 0
					0020	ff f
					0030	00 0

При пинге компьютера из другого VLAN (PC3) ARP-запрос уходит к коммутатору без тега:

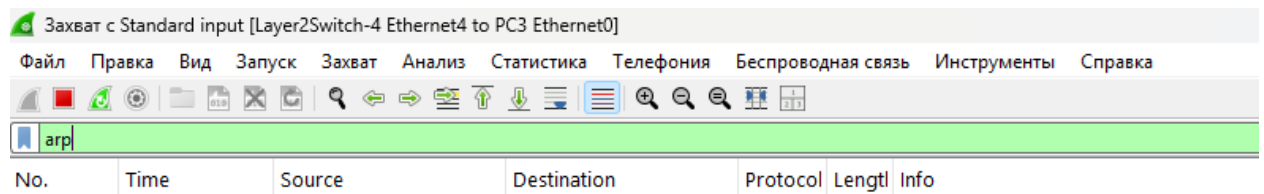
Захват с Standard input [Layer2Switch-3 Ethernet5 to PC2 Ethernet0]						
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка						
Примените фильтр отображения ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Length	Info
217	296.887830	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
218	296.913542	0c:fa:40:2e:00:05	0c:fa:40:2e:00:05	LOOP	60	Reply
219	298.892814	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
220	300.895298	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
221	302.905723	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
222	304.899172	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
223	306.903280	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
224	306.932843	0c:fa:40:2e:00:05	0c:fa:40:2e:00:05	LOOP	60	Reply
225	308.919847	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
226	310.922226	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
227	312.932728	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
228	314.936986	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
229	315.316659	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.3? Tell 192.168.20.2
230	315.708321	0c:fa:40:2e:00:05	CDP/VTP/DTP/PagP/UD...	CDP	424	Device ID: Switch Port ID: GigabitEthernet1/1
231	316.317372	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.3? Tell 192.168.20.2
232	316.942328	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
233	316.942364	0c:fa:40:2e:00:05	0c:fa:40:2e:00:05	LOOP	60	Reply
234	317.317394	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.3? Tell 192.168.20.2
235	318.945674	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
236	320.954504	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
237	321.554630	0c:fa:40:2e:00:05	CDP/VTP/DTP/PagP/UD...	DTP	60	Dynamic Trunk Protocol
238	321.554718	0c:fa:40:2e:00:05	CDP/VTP/DTP/PagP/UD...	DTP	90	Dynamic Trunk Protocol
239	322.958002	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
240	324.971546	0c:fa:40:2e:00:05	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 8 Port = 0x8006
> Frame 229: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0 > Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff) > Address Resolution Protocol (request)						
					0000	ff ff
					0010	08 00
					0020	ff ff
					0030	00 00

Между коммутаторами перемещается без тега:

Захват с Standard input [Layer2Switch-3 Ethernet2 to Layer2Switch-2 Ethernet2]									
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка									
Примените фильтр отображения ... <Ctrl-/>									
No.	Time	Source	Destination	Protocol	Length	Info			
242	280.832638	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
243	282.840148	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
244	284.837149	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
245	286.840689	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
246	286.868037	0c:fa:40:2e:00:02	0c:fa:40:2e:00:02	LOOP	60	Reply			
247	287.386155	0c:88:ae:68:00:02	0c:88:ae:68:00:02	LOOP	60	Reply			
248	288.853585	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
249	290.859928	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
250	292.879402	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
251	293.729042	0c:fa:40:2e:00:02	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol			
252	294.874594	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
253	295.255643	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.3? Tell 192.168.20.2			
254	296.256611	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.3? Tell 192.168.20.2			
255	296.877816	0c:fa:40:2e:00:02	0c:fa:40:2e:00:02	LOOP	60	Reply			
256	296.879984	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
257	297.256219	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.3? Tell 192.168.20.2			
258	297.418241	0c:88:ae:68:00:02	0c:88:ae:68:00:02	LOOP	60	Reply			
259	298.883405	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
260	300.892251	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
261	301.784666	0c:88:ae:68:00:02	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol			
262	302.895772	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
263	304.906546	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
264	306.879003	0c:fa:40:2e:00:02	0c:fa:40:2e:00:02	LOOP	60	Reply			
265	306.912060	0c:88:ae:68:00:02	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8003	
> Frame 253: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0 > Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff) > Address Resolution Protocol (request)									
						0000	ff ff		
						0010	08 00		
						0020	ff ff		
						0030	00 00		

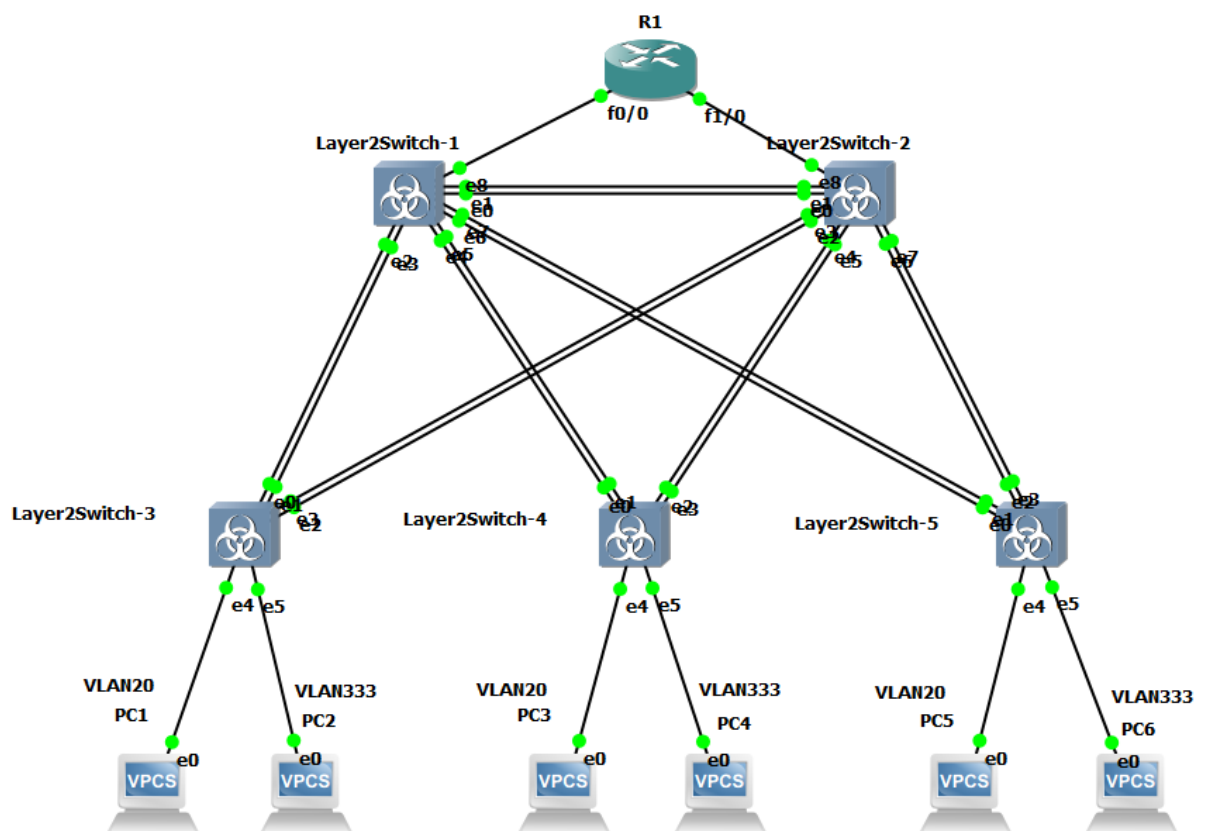
Захват с Standard input [Layer2Switch-2 Ethernet4 to Layer2Switch-4 Ethernet2]									
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка									
Примените фильтр отображения ... <Ctrl-/>									
No.	Time	Source	Destination	Protocol	Length	Info			
190	220.658696	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
191	221.678908	0c:88:ae:68:00:04	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol			
192	222.668263	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
193	223.200700	0c:6b:9e:d5:00:02	0c:6b:9e:d5:00:02	LOOP	60	Reply			
194	224.672458	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
195	225.054522	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.3? Tell 192.168.20.2			
196	225.260975	0c:88:ae:68:00:04	CDP/VTP/DTP/PagP/UD...	CDP	424	Device ID: Switch Port ID: GigabitEthernet1/0			
197	226.060473	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.3? Tell 192.168.20.2			
198	226.677830	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
199	227.054494	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.3? Tell 192.168.20.2			
200	227.219441	0c:88:ae:68:00:04	0c:88:ae:68:00:04	LOOP	60	Reply			
201	228.681245	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
202	230.690104	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
203	232.693662	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
204	233.213151	0c:6b:9e:d5:00:02	0c:6b:9e:d5:00:02	LOOP	60	Reply			
205	234.704476	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
206	236.709973	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
207	237.246483	0c:88:ae:68:00:04	0c:88:ae:68:00:04	LOOP	60	Reply			
208	238.715911	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
209	238.888062	0c:6b:9e:d5:00:02	CDP/VTP/DTP/PagP/UD...	DTP	62	Dynamic Trunk Protocol			
210	240.719027	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
211	242.722136	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
212	243.214640	0c:6b:9e:d5:00:02	0c:6b:9e:d5:00:02	LOOP	60	Reply			
213	244.728096	0c:88:ae:68:00:04	PVST+	STP	68	Conf. Root = 32768/333/0c:5f:1e:17:00:00	Cost = 4	Port = 0x8005	
> Frame 195: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0 > Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff) > Address Resolution Protocol (request)									
						0000	ff ff		
						0010	08 00		
						0020	ff ff		
						0030	00 00		

До PC3 ARP-запрос не доходит и PC2 не получает ответа:



4) Сохранить файлы конфигураций устройств в виде набора файлов с именами, соответствующими именам устройств

5*) Опциональное задание: Добавить в схему маршрутизатор, подключенный к коммутаторам Layer2Switch1 и Layer2Switch2, настроить через него маршрутизацию между VLAN



Настройка роутера:

R1#enable

R1#conf t

Enter configuration commands, one per line. End with CNTL/Z.

R1(config)#int fa0/0

R1(config-if)#ip address 192.168.20.100 255.255.255.0

R1(config-if)#no sh

R1(config-if)#

*Mar 1 00:01:39.471: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up

*Mar 1 00:01:40.471: %LINEPROTO-5-UPDOWN: Line protocol on Interface
FastEthernet0/0, changed state to up

R1(config-if)#exit

R1(config)#int fa1/0

R1(config-if)#ip address 192.168.33.100 255.255.255.0

R1(config-if)#no sh

R1(config-if)#exit

R1(config)#exit

Настройка соединений с роутером на коммутаторах:

Коммутатор 1:

Switch#enable

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#int gi2/0

Switch(config-if)#switchport mode access

Switch(config-if)#switchport access vlan 20

Switch(config-if)#exit

Switch(config)#exit

Коммутатор 2:

```
Switch>enable
```

```
Switch#conf t
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Switch(config)#int gi2/0
```

```
Switch(config-if)#
```

```
*May 13 20:19:59.053: %CDP-4-DUPLEX_MISMATCH: duplex mismatch discovered on  
GigabitEthernet2/0 (not full duplex), with R1 FastEthernet1/0 (full duplex).
```

```
Switch(config-if)#switchport mode access
```

```
Switch(config-if)#switchport access vlan 333
```

```
Switch(config-if)#exit
```

```
Switch(config)#exit
```

Поменяем адреса для VLAN 333 и назначим шлюз по умолчанию:

PC1:

```
PC1> ip 192.168.20.1 24 192.168.20.100
```

Checking for duplicate address...

```
PC1 : 192.168.20.1 255.255.255.0 gateway 192.168.20.100
```

PC2:

```
PC2> ip 192.168.33.32 24 192.168.33.100
```

Checking for duplicate address...

```
PC2 : 192.168.33.32 255.255.255.0 gateway 192.168.33.100
```

PC3:

PC3> ip 192.168.20.3 24 192.168.20.100

Checking for duplicate address...

PC3 : 192.168.20.3 255.255.255.0 gateway 192.168.20.100

PC4:

PC4> ip 192.168.33.34 24 192.168.33.100

Checking for duplicate address...

PC4 : 192.168.33.34 255.255.255.0 gateway 192.168.33.100

PC5:

PC5> ip 192.168.20.5 24 192.168.20.100

Checking for duplicate address...

PC5 : 192.168.20.5 255.255.255.0 gateway 192.168.20.100

PC6:

PC6> ip 192.168.33.36 24 192.168.33.100

Checking for duplicate address...

PC6 : 192.168.33.36 255.255.255.0 gateway 192.168.33.100

Пинг компьютера из другого VLAN (от PC6 к PC1):

PC6> ping 192.168.20.1

192.168.20.1 icmp_seq=1 timeout

84 bytes from 192.168.20.1 icmp_seq=2 ttl=63 time=17.754 ms

84 bytes from 192.168.20.1 icmp_seq=3 ttl=63 time=26.404 ms

84 bytes from 192.168.20.1 icmp_seq=4 ttl=63 time=15.423 ms

84 bytes from 192.168.20.1 icmp_seq=5 ttl=63 time=15.722 ms

Пакеты прошли через маршрутизатор без тегов:

Захват с Standard input [R1 FastEthernet1/0 to Layer2Switch-2 Ethernet8]

ФайлПравкаВидЗапускЗахватАнализСтатистикаТелефонияБеспроводная связьИнструментыСправка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
334	404.101310	0c:88:ae:68:00:08	0c:88:ae:68:00:08	LOOP	60	Reply
335	404.119951	0c:88:ae:68:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8009
336	406.148921	0c:88:ae:68:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8009
337	408.167737	0c:88:ae:68:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8009
338	408.283798	cc:01:10:fd:00:10	cc:01:10:fd:00:10	LOOP	60	Reply
339	410.186300	0c:88:ae:68:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8009
340	410.735471	Private_66:68:05	Broadcast	ARP	64	Who has 192.168.33.100? Tell 192.168.33.36
341	410.749112	cc:01:10:fd:00:10	Private_66:68:05	ARP	60	192.168.33.100 is at cc:01:10:fd:00:10
342	410.764216	192.168.33.36	192.168.20.1	ICMP	98	Echo (ping) request id=0x4fe4, seq=1/256, ttl=64 (no response found!)
343	411.075805	0c:88:ae:68:00:08	CDP/VTP/DTP/PAgP/UD...	CDP	424	Device ID: Switch Port ID: GigabitEthernet2/0
344	412.211913	0c:88:ae:68:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8009
345	412.755030	192.168.33.36	192.168.20.1	ICMP	98	Echo (ping) request id=0x51e4, seq=2/512, ttl=64 (reply in 346)
346	412.771232	192.168.20.1	192.168.33.36	ICMP	98	Echo (ping) reply id=0x51e4, seq=2/512, ttl=63 (request in 345)
347	413.780415	192.168.33.36	192.168.20.1	ICMP	98	Echo (ping) request id=0x52e4, seq=3/768, ttl=64 (reply in 348)
348	413.797644	192.168.20.1	192.168.33.36	ICMP	98	Echo (ping) reply id=0x52e4, seq=3/768, ttl=63 (request in 347)
349	414.197665	0c:88:ae:68:00:08	0c:88:ae:68:00:08	LOOP	60	Reply
350	414.228499	0c:88:ae:68:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8009
351	414.800521	192.168.33.36	192.168.20.1	ICMP	98	Echo (ping) request id=0x53e4, seq=4/1024, ttl=64 (reply in 352)
352	414.813494	192.168.20.1	192.168.33.36	ICMP	98	Echo (ping) reply id=0x53e4, seq=4/1024, ttl=63 (request in 351)
353	415.817291	192.168.33.36	192.168.20.1	ICMP	98	Echo (ping) request id=0x54e4, seq=5/1280, ttl=64 (reply in 354)
354	415.829561	192.168.20.1	192.168.33.36	ICMP	98	Echo (ping) reply id=0x54e4, seq=5/1280, ttl=63 (request in 353)
355	416.248714	0c:88:ae:68:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8009
356	418.276846	cc:01:10:fd:00:10	cc:01:10:fd:00:10	LOOP	60	Reply
357	418.277470	0c:88:ae:68:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/333/0c:5f:1e:17:00:00 Cost = 4 Port = 0x8009

> Frame 340: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0

> Ethernet II, Src: Private_66:68:05 (00:50:79:66:68:05), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

> Address Resolution Protocol (request)

0000 ff ff ff
0010 08 00 06
0020 ff ff ff
0030 00 00 00

Захват с Standard input [Layer2Switch-1 Ethernet8 to R1 FastEthernet0/0]

ФайлПравкаВидЗапускЗахватАнализСтатистикаТелефонияБеспроводная связьИнструментыСправка

Примените фильтр отображения ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
334	406.153338	0c:5f:1e:17:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8009
335	408.165374	0c:5f:1e:17:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8009
336	410.182670	0c:5f:1e:17:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8009
337	411.082744	0c:5f:1e:17:00:08	0c:5f:1e:17:00:08	LOOP	60	Reply
338	412.214441	0c:5f:1e:17:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8009
339	412.473655	0c:5f:1e:17:00:08	CDP/VTP/DTP/PAgP/UD...	CDP	424	Device ID: Switch Port ID: GigabitEthernet2/0
340	414.235203	0c:5f:1e:17:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8009
341	414.325636	cc:01:10:fd:00:00	cc:01:10:fd:00:00	LOOP	60	Reply
342	416.254304	0c:5f:1e:17:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8009
343	416.841293	cc:01:10:fd:00:00	Broadcast	ARP	60	Who has 192.168.20.1? Tell 192.168.20.100
344	416.847245	Private_66:68:00	cc:01:10:fd:00:00	ARP	60	192.168.20.1 is at 00:50:79:66:68:00
345	418.279669	0c:5f:1e:17:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8009
346	418.833168	192.168.33.36	192.168.20.1	ICMP	98	Echo (ping) request id=0x51e4, seq=2/512, ttl=63 (reply in 347)
347	418.834538	192.168.20.1	192.168.33.36	ICMP	98	Echo (ping) reply id=0x51e4, seq=2/512, ttl=64 (request in 346)
348	419.859571	192.168.33.36	192.168.20.1	ICMP	98	Echo (ping) request id=0x52e4, seq=3/768, ttl=63 (reply in 349)
349	419.867923	192.168.20.1	192.168.33.36	ICMP	98	Echo (ping) reply id=0x52e4, seq=3/768, ttl=64 (request in 348)
350	420.296155	0c:5f:1e:17:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8009
351	420.875462	192.168.33.36	192.168.20.1	ICMP	98	Echo (ping) request id=0x53e4, seq=4/1024, ttl=63 (reply in 352)
352	420.881759	192.168.20.1	192.168.33.36	ICMP	98	Echo (ping) reply id=0x53e4, seq=4/1024, ttl=64 (request in 351)
353	421.181632	0c:5f:1e:17:00:08	0c:5f:1e:17:00:08	LOOP	60	Reply
354	421.891535	192.168.33.36	192.168.20.1	ICMP	98	Echo (ping) request id=0x54e4, seq=5/1280, ttl=63 (reply in 355)
355	421.897954	192.168.20.1	192.168.33.36	ICMP	98	Echo (ping) reply id=0x54e4, seq=5/1280, ttl=64 (request in 354)
356	422.317043	0c:5f:1e:17:00:08	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:5f:1e:17:00:00 Cost = 0 Port = 0x8009
357	424.328700	cc:01:10:fd:00:00	cc:01:10:fd:00:00	LOOP	60	Reply

> Frame 343: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

> Ethernet II, Src: cc:01:10:fd:00:00 (cc:01:10:fd:00:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)

> Address Resolution Protocol (request)

0000 ff ff
0010 08 00
0020 00 00
0030 00 00